

Intermediate Macro

Chapter 7: The Labor Market

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- ① Chapter 7: The Labor Market
 - 7.1 A Tour of the Labor Market
 - 7.2 Movements in Unemployment
 - 7.3 Wage Determination
 - 7.4 Price Determination
 - 7.5 The Natural Rate of Unemployment
 - 7.6 Where We Go from Here



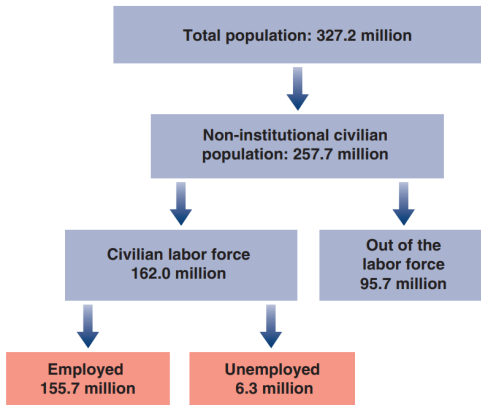
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- Chapter 7 reviews the definition of key labor market terms introduced in Chapter 2 and introduces several new ones, including the **participation rate**, **discouraged workers**, **separations**, **layoffs**, and **quits**.
- Chapter 7 makes use of a **production function**.
- Chapter 7 introduces **wage-setting** and **price-setting** relations.
- Chapter 7 defines analytically the **natural rate of unemployment** and the **natural level of output**.

- Chapter 7 assumes that labor is the only factor of production. This assumption is effective until Chapter 10, which introduces growth.
- We assume a constant returns to scale (CRS) production function with fixed technology. This specification of the production function (labor only, CRS, and fixed technology) implies that the real wage is constant.

7.1 A Tour of the Labor Market

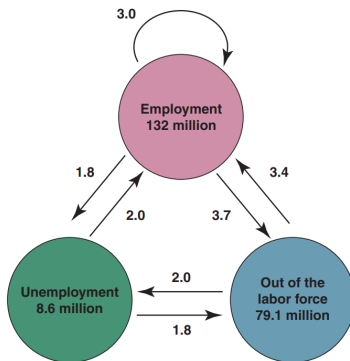
- The U.S. labor market is characterized by large flows between the three states of labor market activity: employed, unemployed, and out of the labor force.



Average Monthly Flows

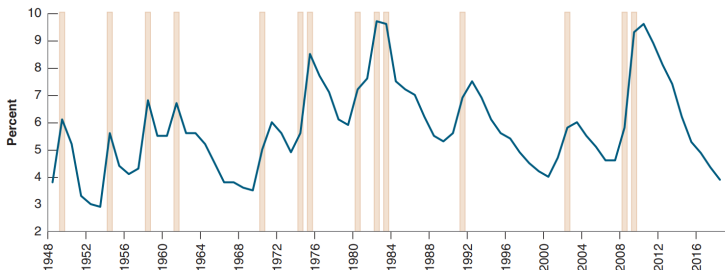
7.1 A Tour of the Labor Market

- Note that unemployment in the United States is not a stagnant pool of workers waiting indefinitely for jobs.



Fluctuations in Unemployment Rate

7.2 Movements in Unemployment

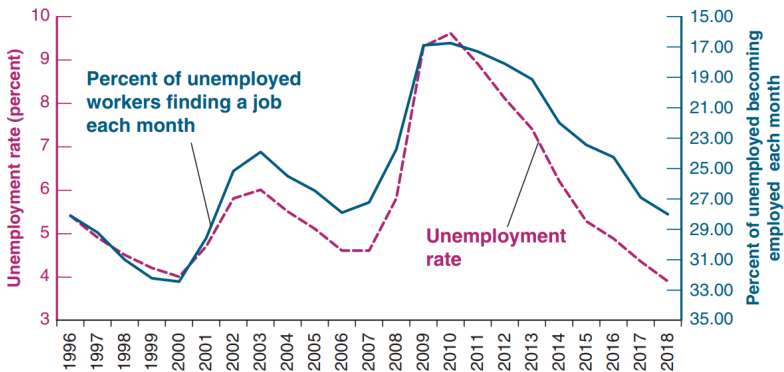


- Two important features about the U.S. unemployment rate:
 - After World War II, there was an upward trend in the unemployment rate until the mid-1980s. Since then, the unemployment rate has declined.
 - Year-to-year fluctuations in the unemployment rate are associated with recessions and expansions.

Implications for Unemployed Workers

7.2 Movements in Unemployment

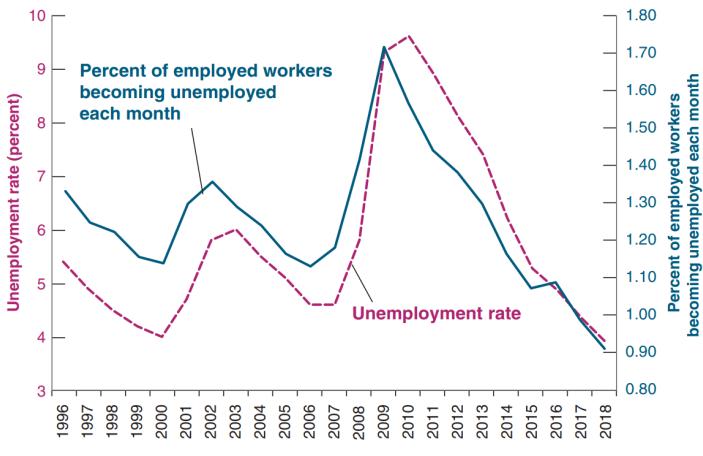
- When the unemployment rate is high, the proportion of unemployed workers finding jobs is low.



Implications for Employed Workers

7.2 Movements in Unemployment

- When the unemployment rate is high, the proportion of employed workers losing their jobs is high.



Bargaining

7.3 Wage Determination

- Wage bargaining between employers and employees takes many forms.
- In some occupations, wages are determined by **collective bargaining** between unions and firms. In the United States, slightly more than 10% of workers are covered by collective bargaining agreements.
- Highly or uniquely skilled workers (e.g., athletes, entertainers) engage in individual bargaining with their employers.
- For jobs that require little skill, employers may make take-it-or-leave-it wage offers.

Efficiency Wages

7.3 Wage Determination

- **Efficiency wage theories** are motivated by the idea that labor productivity is related to the wage.
- Paying a high wage may improve employee morale.
- A high wage may reduce turnover, which can be advantageous to the firm if it takes times to train workers.
- From this perspective, firms have an incentive to offer a wage above the **reservation wage**—the wage at which a worker is indifferent between working or becoming unemployed.

From Henry Ford to Jeff Bezos

7.3 Wage Determination

Table 1 Annual Turnover and Layoff Rates (%) At Ford, 1913–1915

	1913	1914	1915
Turnover rate (%)	370	54	16
Layoff rate (%)	62	7	0.1

- In 1914, Henry Ford, the builder of Model-T, announced that his company would pay all qualified employees a minimum of \$5.00 a day for an eight-hour day, compared to previously an average \$2.30 for a nine-hour day.
- The turnover rate plunged from 350% in 1913 to 16% in 1915.
- Amazon raised wages in a similar manner in October 2018.
- Jeff Bezos announced that Amazon would increase the minimum wage paid to its regular and seasonal workers to \$15 an hour (compared to a federal minimum wage of \$7.50).

Wages, Prices, and Unemployment

7.3 Wage Determination

- We can capture our discussion of wage determination by using the following equation:

$$W = P^e F(u, z) \quad (7.1)$$

(-,+)

where u denotes the unemployment rate and z stands for all other variables that may affect the outcome of wage setting.

- Let us look at each factor in more detail one by one.

The Unemployment Rate

7.3 Wage Determination

- Wage outcomes depend on labor market conditions, which can be proxied by the unemployment rate u .
- When the unemployment rate is high, it is relatively easy for firms to replace workers and difficult for workers to find new jobs, so worker bargaining power is low.
- In addition, workers are highly motivated to work and are unlikely to quit, so the efficiency wage motive is weaker.

The Other Factors

7.3 Wage Determination

- Given the unemployment rate, there are institutional and structural factors (summarized by the variable z) that affect the bargaining power of workers relative to employers.
- These factors include, among other things, the generosity of **unemployment insurance** and the level of the minimum wage.

The Expected Price Level

7.3 Wage Determination

- The nominal wage depends on the price level, because both workers and firms care about the real wage.
- However, wages are changed infrequently, so the price level that matters is the one that prevails over the duration of the wage contract.
- Since the future price level is unknown at the time when the wage contract is set, wage determination depends on the *expected price level*, rather than the *actual price level*.

CRS Production Function

7.4 Price Determination

- Assume that labor is the only factor of production and that firms operate under constant returns to scale.
- We will write the production function as follows:

$$Y = AN$$

where Y is output, A is **labor productivity**, and N is employment.

- Given the assumption that labor productivity A is constant, we can choose the units of output so that one worker produces one unit of output—in other words, so that $A = 1$.
- With this assumption, the **production function** becomes:

$$Y = N \tag{7.2}$$

Marginal Cost of Output and Employment

7.4 Price Determination

- Equation (7.2) implies that the cost of producing one more unit of output is the cost of employing one more worker, at wage W .
- If there was perfect competition in the goods market, the price of a unit of output would be equal to marginal cost: P would be equal to W .
- But many goods markets are not competitive, and firms charge a price higher than their marginal cost.
- A simple way of capturing this fact is to assume that firms set their price according to:

$$P = (1 + m)W \quad (7.3)$$

where m is the **markup** of the price over the cost.

The Wage-Setting Relation

7.5 The Natural Rate of Unemployment

- Given the assumption that nominal wages depend on the actual price level rather than on the expected price level, equation (7.1) becomes:

$$W = PF(u, z)$$

- Dividing both sides by the price level,

$$\frac{W}{P} = F(u, z)_{(-,+)} \quad (7.4)$$

- This relation between the real wage and the rate of unemployment is called the **wage-setting relation**.

The Price-Setting Relation

7.5 The Natural Rate of Unemployment

- If we divide both sides of the price determination equation (7.3) by the nominal wage, we get

$$\frac{P}{W} = (1 + m) \quad (7.5)$$

- When we invert this equation, we get:

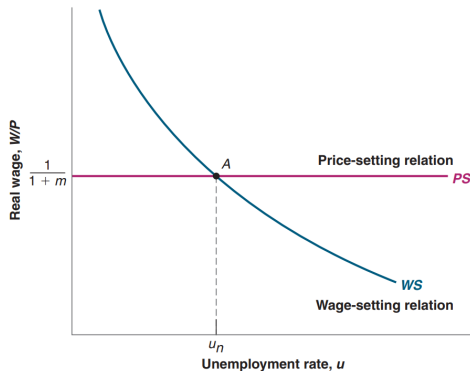
$$\frac{W}{P} = \frac{1}{1 + m} \quad (7.6)$$

- Note what this equation says: *Price-setting decisions determine the real wage paid by firms.*

Equilibrium Real Wages and Unemployment

7.5 The Natural Rate of Unemployment

- Equilibrium in the labor market requires that the real wage chosen in wage setting be equal to the real wage implied by price setting.



The Natural Rate of Unemployment

7.5 The Natural Rate of Unemployment

- We can also characterize the equilibrium unemployment rate algebraically.
- Eliminating W/P between equations (7.4) and (7.6) gives

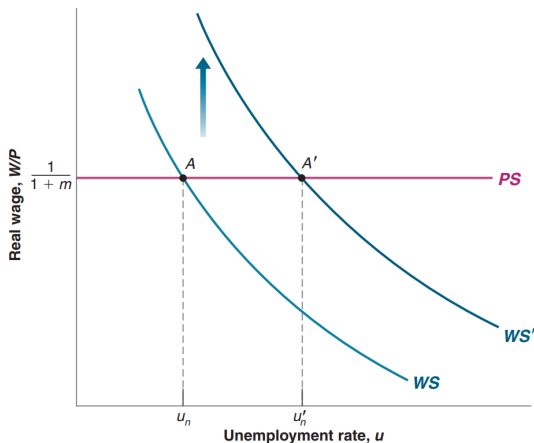
$$F(u, z) = \frac{1}{1 + m} \quad (7.7)$$

- The equilibrium unemployment rate that satisfies equation (7.7) is called the **natural rate of unemployment** u_n .

Unemployment Benefits and u_n

7.5 The Natural Rate of Unemployment

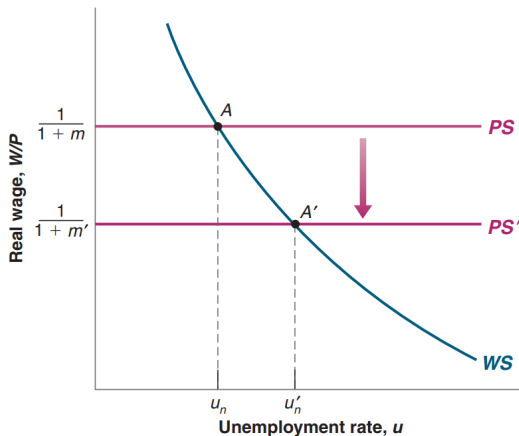
- An increase in unemployment benefits leads to an increase in the natural rate of unemployment.



Markups and u_n

7.5 The Natural Rate of Unemployment

- An increase in the markup leads to an increase in the natural rate of unemployment.



The Medium Run

7.6 Where We Go from Here

- We have assumed that the price level is equal to the expected price level.
- Chapter 7 discusses the determination of the unemployment rate and output in the medium run, a time frame in which it is reasonable to assume that the price level equals the expected price level.
- *In the short run*, when the expected price level need not equal the actual price level, the demand factors discussed in the previous chapters affect the unemployment rate.
- In the short run, the price level may well turn out to be different from what is expected when nominal wages are set, so that unemployment is not necessarily equal to the natural rate or output equal to its natural level.
- The next chapter will relax the assumption that the price level is equal to the expected price level.